

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (AIML/CCE) (Credit System) Degree Examinations

May - June 2023

20AM621 / 20CC621 - BLOCKCHAIN TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain various ideas that supported the invention of Bit coin and block chain with a neat diagram.	10	L*2	1	1
b)	Explain any five types of Consensus mechanism.	10	L2	2	1
2. a)	Define Blockchain. What are the Generic Elements of a block chain? Explain briefly.	10	L2	1	1
b)	Differentiate between Stream cipher and Block cipher.	10	L2	2	1
3. a)	Describe the working of the DAO, DAC, DAS and DApps decentralized organizations.	10	L2	1	1
b)	With suitable diagram, explain how public key cryptography is applied for encryption-decryption process and Digital signature process.	10	L2	2	1
Unit – II					
4. a)	With suitable diagram, explain the Ethereum Virtual Machine operation.	10	L2	3	1
b)	With a simple example program, explain the layout of a solidity program.	10	L2	4	3
5. a)	Explain ANY FOUR precompiled contracts in Ethereum.	10	L2	3	1
b)	With a neat generic model of an oracle and smart contract ecosystem, explain the role and types of oracle in smart contract paradigm.	10	L2	4	3
6. a)	Explain the following: i) Ethereum Stack ii) Recardian Contract	10	L2	3	1
b)	With suitable example, explain following in Solidity. i) Events ii) Inheritance	10	L2	4	3
Unit – III					
7. a)	With a neat runtime architecture of Hyperledger fabric, explain transaction processing flow.	10	L2	5	2
b)	Explain the principles followed in designing any Hyperledger projects.	10	L2	5	2
8. a)	Explain five tools in the Hyperledger project.	10	L2	5	2
b)	Explain advantages of Hyperledger and its family of projects to the industry.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
